Yousef Alaa Awad

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Education

University of Central Florida — Orlando, FL
Bachelor of Science in Computer Engineering with University Honors | GPA: 3.77 — Expected Aug 2027
Relevant Courses: Verification of Digital Systems, HDL in Digital Systems, and Adv. Computer Architecture
Affiliations: IEEE-HKN (Zeta Chi Chapter), IEEE, ACM, Burnett Honors College

Work Experience

Advanced Micro Devices, Inc. — Austin, TX
Incoming Validation/Debug Intern — Sept 2026 – Dec 2026

Lockheed Martin Corporation — Orlando, FL
ASIC/FPGA Design CWEP Intern — Feb 2026 – July 2026

- Built a Formal Property Verification (FPV) flow via Synopsys VC Formal from scratch, authoring 20+ SVAs for bounded model checking, catching 3 critical bugs (illegal state transitions, deadlocks) missed by existing functional tests.
- Extended the flow with Formal Security Verification (FSV/SPV) by authoring security properties for an encryption module, formally verifying the module against possible secret leakage.
- Deployed the FPV/FSV suite in GitLab CI/CD with a modular, parameterized structure, reducing new-module verification setup to a copy-paste testbench and a single pipeline variable toggle.

Embedded Software Engineering CWEP Intern — Apr 2025 – Feb 2026

- Designed a system-level Hardware-in-the-Loop (HIL) simulation environment in C++, enabling high-fidelity testing within the existing workflow for 20+ software engineers by implementing a CIGI v2 compliant image generator.
- Remediated 250+ security violations flagged by static code analysis, resolving critical memory-safety issues including buffer overflows and unsafe pointer access across the codebase.
- Engineered the migration of legacy C++/Qt tooling environments from Visual Studio to a CMake build system, slashing compilation times by 50% and establishing a modular architecture to improve future scalability of the application.

Projects

KnightCore – RISC-V SoC – 2025 AMD Hardware Competition – Group Project — Apr 2025 – Aug 2025

- Developed a coverage-driven verification testbench in Python (Cocotb) to validate a custom 32-bit RISC-V SoC (RV32I ISA), implementing constrained-random stimulus to effectively explore the design state space.
- Developed a comprehensive test suite with directed tests for all ISA-defined instructions and constrained-random stimulus generation to stress the ALU and branch prediction logic, leading to the discovery and resolution of 5 critical RTL bugs.

2025 SouthEastCon Robotics Competition – Embedded Software Lead – Group Project — Sep 2024 – Apr 2025

- Managed Git/GitHub workflows with branching strategies, code reviews, and issue tracking to ensure team-wide consistency. Built CI/CD pipelines for automated testing and deployment, and implemented HIL testing for early hardware validation.
- Led the embedded software pipeline using ROS2, enabling low-latency control between modular hardware and decision nodes, which was then validated via Gazebo/OpenCV, contributing to 1st place in Design and 2nd place in Performance.

CyndaQuil Compiler/Language – Solo Project — Sep 2024 – Jan 2025

- Designed and implemented a statically-typed compiled language from scratch in C++, building the full toolchain (parser, semantic analyzer, and x86 codegen) with CMake managing a multi-stage build pipeline.
- Engineered low-level system components and optimized performance, integrating Assembly and C to enhance hardware-software interfacing, ensuring minimal latency and efficient computation for embedded systems.

Leadership

Eta Kappa Nu (IEEE-HKN)
Chapter President, Zeta Chi Chapter — Apr 2026 – Present

- Directing chapter operations following Key Chapter recognition by expanding induction efforts, building an active event calendar and a more engaged membership base, and re-engaging alumni to strengthen long-term chapter sustainability.

Institute of Electrical and Electronics Engineers
Treasurer, UCF Student Branch — Apr 2025 – Apr 2026

- Managed a $10,000+ budget and facilitated $6,500+ in sponsorship funding to support student project development, directly enabling the Student Branch's 2nd place finish in the Circuit Design Competition at IEEE Region 3's SouthEastCon.

Technical Skills

Formal Verification: Synopsys VC Formal, SystemVerilog Assertions (SVA), Bounded Model Checking, Formal Property Verification (FPV), Formal Security Verification (FSV/SPV)
Languages: SystemVerilog, Verilog, C++, Tcl, Python, C, Bash, Rust, Assembly (x86, RISC-V, MIPS)
Computer Architecture: RISC-V ISA, AXI/Memory-Mapped Interfaces, Pipelining, Caching, Memory Hierarchy
Hardware & EDA Tools: Synopsys VC Formal, Synopsys VCS, Xilinx Vivado, Verilator
Systems & Scripting: Linux, Git, GitLab CI/CD, ROS/ROS2, Real-Time C++, CMake/Make